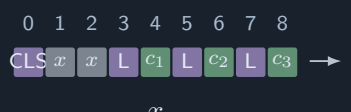
$$q = f_{text}(H_{CLS,:}), \quad L = f_{label}(H[P_L, :]), \quad s = qL^{\mathsf{T}}, \quad p = \sigma(s)$$

1. Token layout

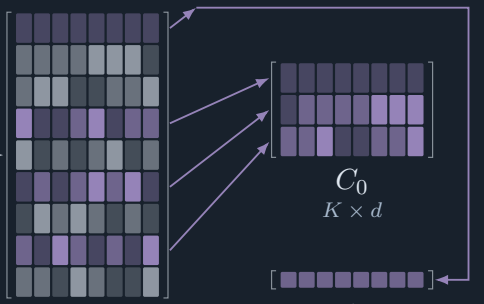


x
T integer token IDs

2. GLiClass

DeBERTa-v3-small
6 bidirectional
blocks; d=768
uni-encoder /
simple scorer

3. Contextual states and indices



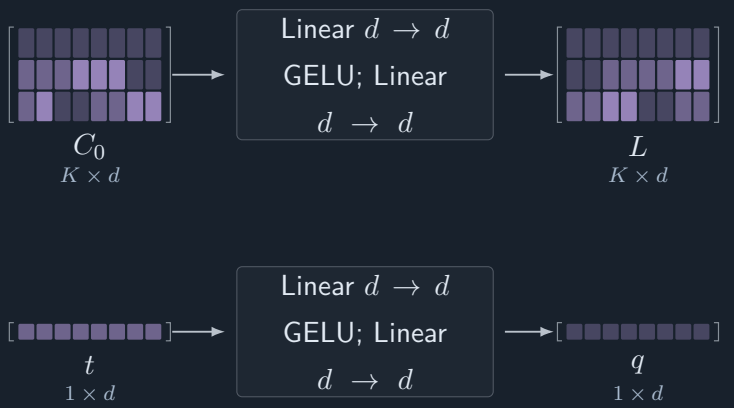
H
$T \times d$

C_0
$K \times d$

t
$1 \times d$

Joint bidirectional input
read CLS + label markers
no feature L2;
no temperature

4. Distinct text and label projectors



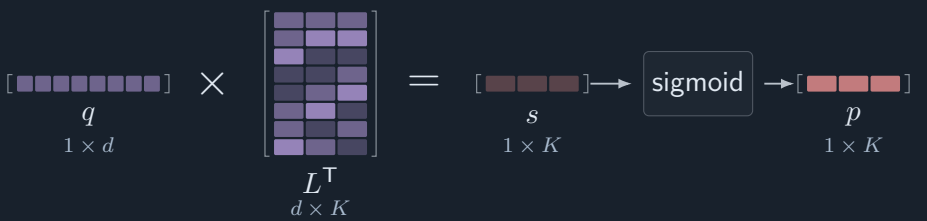
C_0
$K \times d$

L
$K \times d$

t
$1 \times d$

q
$1 \times d$

5. Text-label dot product



q
$1 \times d$

L^{T}
$d \times K$

s
$1 \times K$

p
$1 \times K$

Axes $d = 768$; Single-question illustration; batch omitted. Tile counts encode axes, not literal sizes.

Objects Gray: input/context; violet: hidden states; orange: parameter matrices; red: logits/probabilities. Positions, code IDs and owner maps are integers, not scores. Tiles show roles, not measured magnitudes.

Mechanism CLS and label markers see the joint input. Independent d-to-d-to-d projectors give a text/label dot product; per-label sigmoid does not imply independent pairs.

Named multi-label loss branch is BCE; the old checkpoint training history is not recovered. The selected config has no feature L2 or temperature scale.